 Exam #77777

Department of Physics
University of Florida

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PHY 2049 Final Exam April 30, 2024

On my honor, I have neither given nor received unauthorized aid on this examination

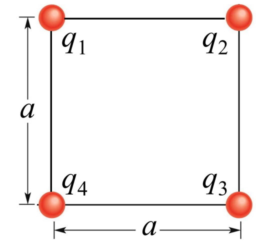
Name
Last: ____________ First: ____________ Signature: ____________

Directions

1. Print your name on this question sheet and sign it.
2. Code your 5-digit exam code (at the top of this question sheet) on your Scantron in columns 76–80.
3. Code your name and 8-digit UFID on the Scantron. **Darken all circles completely on the Scantron** using a #2 pencil or **dark blue** or **black** ink. Do not make any stray marks or the Scantron may not read properly.
4. Do all scratch work anywhere on this question sheet that you like. At the end of the exam, the question sheet and Scantron are both to be turned in. *No credit will be given without both Scantron sheet and question sheet.*
5. The answers are rounded, so choose the closest to exact. There is no penalty for guessing.

Correct answer marked with ♣

1. Four point charges, $q_1 = q_2 = 1\ \mu$C and $q_3 = q_4 = -2\ \mu$C, are located at the corners of a square with side length $a = 10$ cm, as shown in the figure. What is the magnitude of the horizontal component of the force on q_1, in newtons?



(1) ♣ 0.26
(2) 0.43
(3) 0.67
(4) 0.84
(5) 1.12

Solution: q_1 is repelled from q_2 producing a force in the negative x-direction. q_1 is attracted to q_3 producing a positive x-component of the force on q_1. The force between q_1 and q_4 is in the y-direction. The net x-component of the force on q_1 is

$$F_x = -\frac{k(1\,\mu C)(1\,\mu C)}{(10\,cm)^2} + \frac{k(1\,\mu C)(2\,\mu C)}{(\sqrt{2}\ 10\,cm)^2}\cos(45°) = -0.26\,N.$$

The magnitude of the horizontal component of the force on q_1 is thus **0.26 N**.

2. A proton is shot in the $\hat{\imath}$ direction with initial speed 10 km/s in a uniform electric field, which points in the $-\hat{\imath}$ direction. The particle travels 1.0 m, then turns around and travels in the $-\hat{\imath}$ direction. What is the magnitude of the electric field, in N/C? The mass of a proton is 1.67×10^{-27} kg.

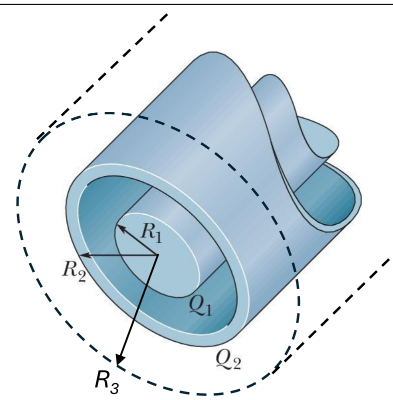
(1) ♣ 0.52 (2) 0.13 (3) 1.7 (4) 4.8 (5) 14

Solution: Using the one dimensional kinematics equations,

$$\begin{aligned} x &= x_o + v_o t - \frac{1}{2}at^2 \\ v &= v_o - at, \end{aligned}$$

with $x_o = 0$, $x = 1$ m, $v_o = 10^4$ m/s, and $v = 0$, we find that $x = v_o^2/(2a)$ or $a = v_o^2/(2x) = 5 \times 10^7$ m/s. Since the magnitude of a is eE/m_p, $E = m_p a/e = \mathbf{0.52\ N/C}$.

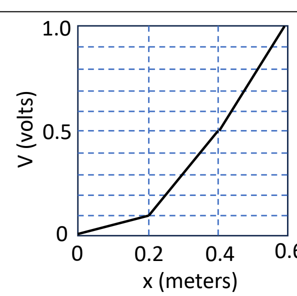
3. The figure is a section of a conducting rod of radius $R_1 = 1$ mm and length $L = 15$ m inside a thin-walled coaxial conducting cylindrical shell of radius $R_2 = 2$ mm and the same length L. The net charge on the rod is $Q_1 = 3$ pC, and the net charge on the shell is $Q_2 = -1$ pC. What is the net electric flux in V·m through the Gaussian surface in the form of a cylinder of length $L = 15$ m and radius $R_3 = 3$ mm that encloses the conductors (dashed line in figure)?



(1) ♣ 0.23
(2) 0.45
(3) 0.57
(4) 0.35
(5) 0.14

Solution: The flux is the charge enclosed divided by ϵ_o: $\Phi = Q_{enc}/\epsilon_o$. The charge enclosed is $3\ pC - 1\ pC = 2\ pC$ so $\mathbf{\Phi = 0.23\ V \cdot m}$.

4. The electric potential as a function of x is shown in the figure. What is the electric field in the x-direction, E_x, at $x = 0.3$ m?



(1) ♣ -2 V/m
(2) 2 V/m
(3) -1.5 V/m
(4) 1.5 V/m
(5) -4 V/m

Solution: $E_x = -\partial V/\partial x$ is the negative of the slope in the graph: $E_x = -0.4V/0.2m = \mathbf{-2\ V/m}$.

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5. An electron moves in the electrical potential created by a proton: $V(r) = k|e|/r$. Since the proton is much more massive than the electron, treat the proton as fixed in this problem. If the electron is initially at rest 1.0 cm from the proton, what is its speed in m/s when it is 0.5 cm from the proton? The mass of an electron is 9.1×10^{-31} kg.

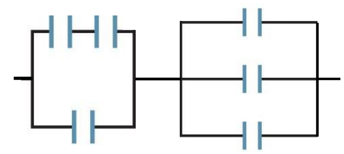
(1) ♣ 225 (2) 318 (3) 159 (4) 403 (5) 550

Solution: This is a conservation of energy problem:

$$0 - \frac{ke^2}{1\ cm} = \frac{1}{2}m_e v^2 - \frac{ke^2}{0.5\ cm}.$$

Solve to find $\mathbf{v} = \mathbf{225\ m/s}$.

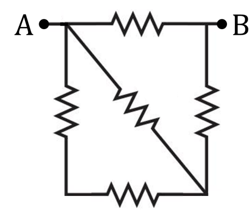
6. In the figure at right all the capacitors are 1 μF. What is the equivalent capacitance of the circuit in μF?



(1) ♣ 1.0 (2) 4.5 (3) 3.0 (4) 1.5 (5) 0.67

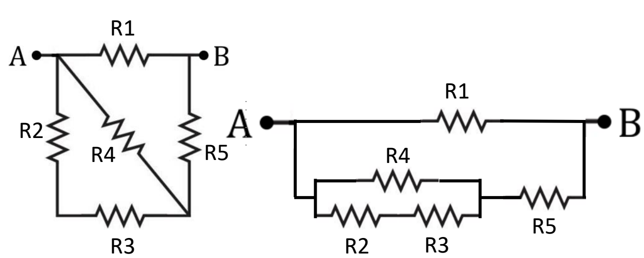
Solution: The first three capacitors on the left have equivalent capacitance of 1.5 μF. The three capacitors on the right have equivalent capacitance of 3 μF. Add these two sets of capacitors in series to get **1** μ**F**.

7. In the figure, all resistances are 400 Ω. Find the equivalent resistance between points A and B in ohms.



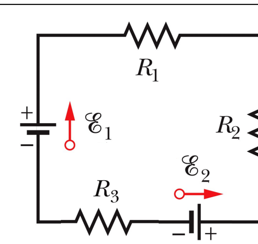
(1) ♣ 250
(2) 150
(3) 350
(4) 450
(5) 550

Solution: The hardest part of this problem is to redraw the circuit so that it is clear which resistors are in series and which are in parallel. This is done below.



The equivalent resistance is then **250Ω**.

8. In the circuit shown, $\mathcal{E}_1 = 10$ V, $\mathcal{E}_2 = 20$ V, $R_1 = R_3 = 200$ Ω, and $R_2 = 100$ Ω. What is the voltage (in volts) across R_3?



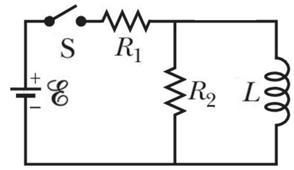
(1) ♣ 4
(2) 6
(3) 10
(4) 15
(5) 0

Solution: Go clockwise around the circuit computing the voltage changes:

$$0 = \mathcal{E}_1 - iR_1 - iR_2 - \mathcal{E}_2 - iR_3.$$

This leads to $i = -0.02$ A clockwise or 0.02 A counterclockwise. The voltage across R_3 is $(0.02\ A)(200\ \Omega) = \mathbf{4\ V}$.

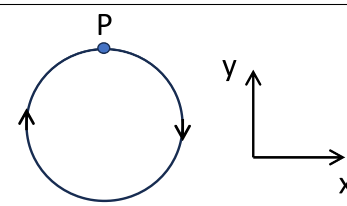
9. In the figure, $\mathcal{E} = 6.0$ V, $R_1 = 20\ \Omega$, $R_2 = 10\ \Omega$, and $L = 20$ mH. There is no current anywhere in the circuit while the switch is open. Immediately after the switch is closed, what is the voltage (in volts) across resistor R_1?



(1) ♣ 4.0 (2) 2.0 (3) 3.0 (4) 6.0 (5) 0.0

Solution: Immediately after the switch is closed the current through the inductor is zero because it can not change instantaneously. This means that the current flows through R_1 and then R_2. The current through R_1 and also R_2 is $6\ V/(30\ \Omega) = 0.2$ A, and the voltage across R_1 is $(0.2\ A)(20\ \Omega) = \mathbf{4\ V}$.

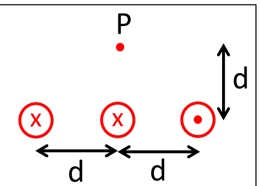
10. A negatively charged particle goes in a circle perpendicular to a magnetic field. What is the direction of the magnetic field and the direction of the force at point P?



(1) ♣ B into page, force in $-\hat{j}$
(2) B out of page, force in $-\hat{j}$
(3) B out of page, force in $\hat{j}$
(4) B into page, force in $\hat{j}$
(5) B out of page, force in $\hat{i}$

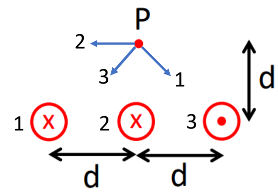
Solution: For circular motion the force is always towards the center of the circle, which is $-\hat{\mathbf{j}}$ here. To get this force for v in the x-direction and a negatively charged particle, the magnetic field must be magnetic field must be **into the page**.

11. Three wires have current either into or out of the page as indicated in the figure. They each carry 3 A of current, and they are separated by $d = 2$ cm. What is the magnitude of the magnetic field at point P in μT?



(1) ♣ 42 (2) 30 (3) 84 (4) 120 (5) 90

Solution: The direction of the magnetic fields due to each wire is shown in the figure below.



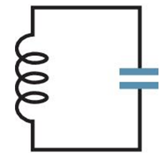
Wires 1 and 3 produce a net magnetic field in the negative y-direction of magnitude $\sqrt{2}(\mu_o i)/(2\pi\sqrt{2}d) = (\mu_o i)/(2\pi d)$. Wire 2 produces a magnetic field in the negative x-direction with magnitude $(\mu_o i)/(2\pi d)$. To get the net magnetic field evaluate $\sqrt{B_x^2 + B_y^2} = \sqrt{2}(\mu_o i)/(2\pi d) = \mathbf{42}\ \mu\mathbf{T}$.

12. The magnetic field is coming out of the page with magnitude $B(t) = 10 - 2t^2$, where B is measured in Tesla and time, t, is measured in seconds. A circular conducting loop with radius 5 cm is placed in the plane of the page. At $t = 2$ seconds what is the induced emf in the loop in mV?

(1) ♣ 63 (2) 16 (3) 31 (4) 20 (5) 54

Solution: From Faraday's law $\mathcal{E} = -d\Phi/dt$. The flux is $\Phi = BA$ so $\mathcal{E} = -A(dB/dt) = \pi r^2(4t)$. Using $t = 2$ seconds and $r = 0.05$ m, we obtain $\mathcal{E} = 0.063\ V = \mathbf{63\ mV}$.

13. A 10 μF capacitor has an initial charge of 100 μC. At time $t = 0$, the capacitor is connected to a 4.0 mH inductor, as shown in the figure. At a later time, it is found that the charge on the capacitor is 60 μC. What is the current (in amperes) in the circuit at that moment?



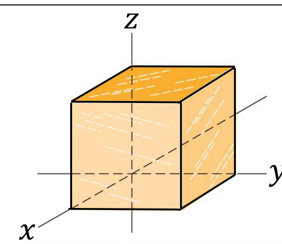
(1) ♣ 0.4 (2) 0.5 (3) 0.3 (4) 0.2 (5) 0.1

Solution: Use conservation of energy for this problem:

$$E = \frac{Q^2}{2C} + \frac{1}{2}Li^2 = \frac{(100\ \mu C)^2}{2(10\ \mu F)} + 0 = \frac{(60\ \mu C)^2}{2(10\ \mu F)} + \frac{1}{2}(4\ mH)i^2.$$

Solve to find $\mathbf{i = 0.4\ A}$.

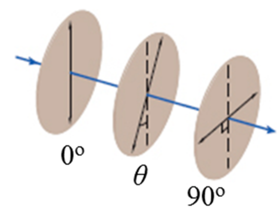
14. A Gaussian cube of edge length 1.0 m is placed in a magnetic field as shown, such that three edges are on the x, y, and z axes and one corner at the origin. On the top face, at $z = 1.0$ m, the magnetic field is given by $\vec{B} = 4\hat{j} + 5\hat{k}$, whereas on the bottom face, at $z = 0$, the magnetic field is given by $\vec{B} = 3\hat{j} + 3\hat{k}$. Here the magnitude of the magnetic field is in tesla. Find the total magnetic flux (in T·m^2) through the four side faces of the cube.



(1) ♣ −2 (2) 2 (3) −1 (4) 1 (5) 0

Solution: The flux through each face is $\vec{B} \cdot \hat{n}A$, where $\hat{n}$ is the normal to surface pointing outwards. The flux through the top face is $5\ T \cdot m^2$. For the bottom face the normal is $-\hat{k}$ so the flux is $-3\ T \cdot m^2$. Because the sum of the magnetic flux through all the surfaces is zero, the flux through the remaining surfaces must be $\mathbf{-2\ T \cdot m^2}$.

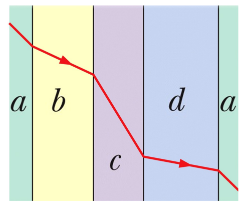
15. Initially unpolarized light is sent into a system of three polarizing sheets whose polarizing directions make angles of 0°, $\theta = 30°$, and 90° with the vertical direction. What fraction of the initial intensity is transmitted by the system?



(1) ♣ 0.094 (2) 0.217 (3) 0.132 (4) 0.176 (5) 0

Solution: After passing through the first sheet, the intensity is reduced by a factor of 2. Each subsequent polarizer reduces the intensity by $\cos(\theta)$, where θ is the angle between adjacent polarizers. $I = 0.5\cos^2(30°)\cos^2(60°)I_o = \mathbf{0.094}I_o$.

16. Light travels through layers of transparent slabs, as shown. The indices of refraction of the slabs are $n_a = 1.2$, $n_b = 1.3$, $n_c = 1.1$, and $n_d = 1.4$. If the angle of incidence at the ab boundary is 45°, what is the angle of incidence at the cd boundary?



(1) ♣ 50.5° (2) 55.3° (3) 60.2° (4) 65.4° (5) 70.6°

Solution: Because the interfaces are parallel, the angle of refraction for one interface is equal to the angle of incidence for the next. In particular this implies that $n_a\sin(\theta_a) = n_b\sin(\theta_b) = n_c\sin(\theta_c)$. Using $\theta_a = 45°$, we find that $\theta_c = \mathbf{50.5°}$.

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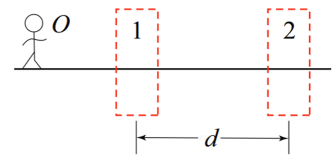
17. A small object is located on the central axis of a spherical convex mirror, whose radius of curvature is 80 cm. If the object is a distance 40 cm from the mirror, how far is the image from the mirror and is the image real or virtual?

(1) ♣ 20 cm, virtual
(2) 20 cm, real
(3) 80 cm, virtual
(4) 80 cm, real
(5) infinity, virtual

Solution: The focal length is -40 cm, and the object distance is 40 cm. This implies that the image is at -20 cm. It is **20 cm** behind the mirror and **virtual**.

18. In the figure, O (the object) stands on the common central axis of a converging lens in region 1 and a diverging lens in region 2. The magnitude of the focal length of lens 1 is 4 cm, and the magnitude of the focal length of lens 2 is 5 cm. The object is 5 cm from lens 1, and the separation between the lenses is $d = 10$ cm. What is the magnitude of the magnification of the image?



(1) ♣ 4 (2) 1 (3) 5 (4) 2 (5) 8

Solution: Because $f_1 = 4$ cm and $p_1 = 5$ cm, the image of the first lens is at $i_1 = 20$ cm. This is 10 cm to the right of the second lens so $p_2 = -10$ cm. Using $f_2 = -5$ cm, we find that $i_2 = -10$ cm. The magnification is $(-i_1/p_1)(-i_2/p_2) = \mathbf{4}$.

19. Light of wavelength 500 nm is sent through two narrow slits onto a screen that is 2 m away. On the screen the separation between the central bright spot and the next bright spot is 1 cm. What is the separation between the slits?

(1) ♣ 100 μm (2) 500 μm (3) 10 μm (4) 50 μm (5) 1 μm

Solution: Use the approximation $\sin(\theta) \approx \theta \approx \tan(\theta)$ for small angles in radians. The angle of the first bright spot is $\tan(\theta) = (1\ cm)(2\ m) = 0.005 \approx \theta$. At this angle $d\sin(\theta) = 1\lambda$. Thus, $d = \lambda/\theta = (500\ nm)/(0.005) = 10^{-4}\ m = \mathbf{100\ \mu m}$.

20. The distance between the sixth minimum and the central maximum of a single-slit diffraction pattern is 2.0 cm with the screen 1.0 m away from the slit, when light of wavelength 700 nm is used. What will be the distance (in cm) between the sixth minimum and the central maximum if the wavelength is 400 nm?

(1) ♣ 1.1 (2) 1.6 (3) 2.0 (4) 2.7 (5) 3.5

Solution: Again use the approximation $\sin(\theta) \approx \theta \approx \tan(\theta)$ for small angles in radians. The sixth minimum occurs at $a\sin(\theta) = 6\lambda \approx a\theta$ with $\theta \approx (2\ cm)/(1\ m) = 0.02$ and $\lambda = 700$ nm. If we decrease the wavelength by 4/7, then θ will decrease by 4/7 and so will the position on the screen: $(4/7)(2\ cm) = 8/7\ cm \approx \mathbf{1.1\ cm}$.

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Scratch paper

Scratch paper

Scratch paper

Formulas – page 1

Constants: $\mu_0 = 4\pi \times 10^{-7}\ \mathrm{T \cdot m/A}$ $\quad \varepsilon_0 = 8.85 \times 10^{-12}\ \mathrm{C^2/N \cdot m^2}$.

$e = 1.6 \times 10^{-19}\ \mathrm{C}$ $\quad k = \frac{1}{4\pi\varepsilon_0} = 8.99 \times 10^9\ \mathrm{N \cdot m^2/C^2}$.

$c = 3.0 \times 10^8\ \mathrm{m/s}$

kilo (k) = 10^3; centi (c) = 10^{-2}; milli (m) = 10^{-3}; micro (μ) = 10^{-6}; nano (n) = 10^{-9}; pico (p) = 10^{-12}

Coulomb's Law: $$F = \frac{1}{4\pi\varepsilon_0}\frac{|q_1||q_2|}{r^2}$$

Electric field:

$\vec{F} = q\vec{E}$ $\qquad E = \frac{qz}{4\pi\varepsilon_0(z^2 + R^2)^{3/2}}$ (charged ring)

$E = \frac{1}{4\pi\varepsilon_0}\frac{|q|}{r^2}$ (point charge) $\qquad E = \frac{\sigma}{2\varepsilon_0}\left(1 - \frac{z}{\sqrt{z^2 + R^2}}\right)$ (charged disk)

$dE = \frac{1}{4\pi\varepsilon_0}\frac{dq}{r^2}$ (continuous) $\qquad E = \frac{\sigma}{2\varepsilon_0}$ (infinite sheet)

Dipoles: $p = qd$ $\qquad \tau = pE\sin\theta$ $\qquad U = -pE\cos\theta$

Gauss' Law:

$\Phi = \int \vec{E}\cdot d\vec{A} = (E\cos\theta)A$ $\qquad E = \frac{\sigma}{\varepsilon_0}$ (conducting surface)

$\varepsilon_0\Phi = \varepsilon_0 \oint \vec{E}\cdot d\vec{A} = q_{\text{enc}}$ $\qquad E = \frac{\lambda}{2\pi\varepsilon_0 r}$ (line of charge)

Electric potential:

$U = qV$ $\qquad V = \frac{1}{4\pi\varepsilon_0}\frac{q}{r}$ (point charge)

$V_f - V_i = -\int_i^f \vec{E}\cdot d\vec{s}$ $\qquad dV = \frac{1}{4\pi\varepsilon_0}\frac{dq}{r}$ (continuous)

$E_x = -\frac{\partial V}{\partial x};\quad E_y = -\frac{\partial V}{\partial y};\quad E_z = -\frac{\partial V}{\partial z}.$ $\qquad U = \frac{1}{4\pi\varepsilon_0}\frac{q_1 q_2}{r}$ (two-particle system)

Capacitors:

$q = CV$ $\qquad C = \frac{\varepsilon_0 A}{d}$ (parallel plate)

$C_{\text{eq}} = \sum_{j=1}^{n} C_j$ (n capacitors in parallel) $\qquad C = 2\pi\varepsilon_0\frac{L}{\ln\left(\frac{b}{a}\right)}$ (cylindrical)

$\frac{1}{C_{\text{eq}}} = \sum_{j=1}^{n}\frac{1}{C_j}$ (n capacitors in series) $\qquad C = 4\pi\varepsilon_0\frac{ab}{b-a}$ (spherical)

$U_E = \frac{q^2}{2C} = \frac{1}{2}CV^2,$ $\qquad C = 4\pi\varepsilon_0 R$ (isolated sphere)

$u_E = \frac{1}{2}\varepsilon_0 E^2.$ $\qquad C = \kappa\frac{\varepsilon_0 A}{d}$ (parallel plate with dielectric)

Formulas – page 2

Current: $i = \frac{dq}{dt}$, $i = \int \vec{J} \cdot d\vec{A}$, $\vec{J} = (ne)\vec{v}_d$,

Resistors: $R = \frac{V}{i}$, $v_R = iR.$ $R = \rho \frac{L}{A}$, $\rho = \frac{1}{\sigma} = \frac{E}{J}$, $P = iV = i^2 R = \frac{V^2}{R}$.

$\frac{1}{R_{eq}} = \sum_{j=1}^{n} \frac{1}{R_j}$ (n resistances in parallel). $R_{eq} = \sum_{j=1}^{n} R_j$ (n resistances in series).

Inductors: $L = \frac{N\Phi_B}{i}$ $\mathscr{E}_L = -L \frac{di}{dt}$. $\frac{L}{l} = \mu_0 n^2 A$ (solenoid).

Inductors add in series and parallel with the same rules as resistors. $\mathscr{E}_2 = -M \frac{di_1}{dt}$

$U_B = \frac{Li^2}{2}$, $u_B = \frac{B^2}{2\mu_0}$ $\mathscr{E}_1 = -M \frac{di_2}{dt}$,

RC circuit: $q = C\mathscr{E}(1 - e^{-t/\tau_C})$. (charging) $q = q_0 e^{-t/\tau_C}$. (discharging) $\tau_C = RC$.

RL circuit: $i = \frac{\mathscr{E}}{R}(1 - e^{-t/\tau_L})$ (rise of current). $i = i_0 e^{-t/\tau_L}$ (decay of current). $\tau_L = \frac{L}{R}$

LC circuit: $q = Q\cos(\omega t + \phi)$ (charge), $i = \frac{dq}{dt} = -\omega Q \sin(\omega t + \phi)$ (current). $\omega = \frac{1}{\sqrt{LC}}$.

RLC circuit: (not driven) $q = Qe^{-Rt/2L}\cos(\omega' t + \phi)$, $\omega' = \sqrt{\omega^2 - (R/2L)^2}$.

Driven circuits: $\mathscr{E} = \mathscr{E}_m \sin \omega_d t$.

One circuit element: R $i_R = \frac{v_R}{R} = \frac{V_R}{R}\sin \omega_d t$.

$X_C = \frac{1}{\omega_d C}$ (capacitive reactance). $i_C = \left(\frac{V_C}{X_C}\right)\sin(\omega_d t + 90°)$.

$X_L = \omega_d L$ (inductive reactance). $i_L = \left(\frac{V_L}{X_L}\right)\sin(\omega_d t - 90°)$.

Series RLC: $Z = \sqrt{R^2 + (X_L - X_C)^2}$ (impedance) $i = I\sin(\omega_d t - \phi)$,

$\tan\phi = \frac{X_L - X_C}{R}$ (phase constant). $I = \frac{\mathscr{E}_m}{\sqrt{R^2 + (X_L - X_C)^2}}$.

Power: $I_{rms} = \frac{I}{\sqrt{2}}$ $\mathscr{E}_{rms} = \frac{\mathscr{E}_m}{\sqrt{2}}$ $P_{avg} = \mathscr{E}_{rms} I_{rms} \cos\phi$

Transformers: $V_s = V_p \frac{N_s}{N_p}$ $I_s = I_p \frac{N_p}{N_s}$ $R_{eq} = \left(\frac{N_p}{N_s}\right)^2 R$,

Formulas – page 3

Forces: $\vec{F} = q\vec{E}$, $\vec{F}_B = q\vec{v}\times\vec{B}$; $F_B = |q|vB\sin\phi$, $\vec{F}_B = i\vec{L}\times\vec{B}$. $F_{ba} = \dfrac{\mu_0 L i_a i_b}{2\pi d}$

Circular motion: $|q|vB = \dfrac{mv^2}{r}$, $r = \dfrac{mv}{|q|B}$. $f = \dfrac{\omega}{2\pi} = \dfrac{1}{T} = \dfrac{|q|B}{2\pi m}$. Hall effect: $n = \dfrac{Bi}{Vle}$,

Magnetic moment: $\mu = NiA$ $\vec{\tau} = \vec{\mu}\times\vec{B}$. $\tau = NiAB\sin\theta$, $U(\theta) = -\vec{\mu}\cdot\vec{B}$.

Magnetic field: $d\vec{B} = \dfrac{\mu_0}{4\pi}\dfrac{i\,d\vec{s}\times\hat{r}}{r^2}$ (Biot–Savart law) $B = \dfrac{\mu_0 i}{2\pi R}$ (long straight wire)

$\oint \vec{B}\cdot d\vec{s} = \mu_0\, i_{enc}$ (Ampere's law) $B = \dfrac{\mu_0 i\phi}{4\pi R}$ (at center of circular arc)

$B = \dfrac{\mu_0 i(2\pi)}{4\pi R} = \dfrac{\mu_0 i}{2R}$ (at center of full circle)

$B = \mu_0 i n$ (ideal solenoid), $B = \dfrac{\mu_0 i N}{2\pi}\dfrac{1}{r}$ (toroid)

Faraday's law: $\mathscr{E} = \oint \vec{E}\cdot d\vec{s}$. $\mathscr{E} = -\dfrac{d\Phi_B}{dt}$ $\Phi_B = \int \vec{B}\cdot d\vec{A}$

Maxwell's eqs.:

Gauss' law for electricity $\oint \vec{E}\cdot d\vec{A} = q_{enc}/\varepsilon_0$

Gauss' law for magnetism $\oint \vec{B}\cdot d\vec{A} = 0$

Faraday's law $\oint \vec{E}\cdot d\vec{s} = -\dfrac{d\Phi_B}{dt}$

Ampere–Maxwell law $\oint \vec{B}\cdot d\vec{s} = \mu_0\varepsilon_0\dfrac{d\Phi_E}{dt} + \mu_0 i_{enc}$ $i_d = \varepsilon_0\dfrac{d\Phi_E}{dt}$.

Polarizers: $I = \frac{1}{2}I_0$. $I = I_0\cos^2\theta$.

Refraction: $n_2\sin\theta_2 = n_1\sin\theta_1$, $\theta_c = \sin^{-1}\dfrac{n_2}{n_1}$ (critical angle).

Mirrors: $i = -p$ (plane mirror). $\dfrac{1}{p} + \dfrac{1}{i} = \dfrac{1}{f} = \dfrac{2}{r}$. $m = -\dfrac{i}{p}$.

Lenses: $\dfrac{1}{f} = \dfrac{1}{p} + \dfrac{1}{i}$ (thin lens), $m = -\dfrac{i}{p}$ (magnification),

Two slits: $d\sin\theta = m\lambda$, for $m = 0, 1, 2, \ldots$ (maxima—bright fringes), $d\sin\theta = (m+\frac{1}{2})\lambda$, for $m = 0,1,2,\ldots$ (minima—dark fringes),

Single slit: $a\sin\theta = m\lambda$, for $m = 1, 2, 3, \ldots$ (minima). $I(\theta) = I_m\left(\dfrac{\sin\alpha}{\alpha}\right)^2$, where $\alpha = \dfrac{\pi a}{\lambda}\sin\theta$

Details of Master Exam Questions

Total of 20 QQ, 0 CQ, 0 RQ questions (0 distinct RQ questions)

* Total of 20 questions defined (20 for each student exam)
* Total weight is 20.00 for each student exam
* Total of 17 figures defined

Q	Type	aType	Wght	#fig	#ans	Shuffle ans?	Correct	**Short text**
1	QQ	Single	1.00	1	5	Yes	1	Four point charges, $q_1 = q_2 = 1$ μC and $q_3 = q_4 = -2$
2	QQ	Single	1.00	0	5	Yes	1	A proton is shot in the $\hat{i}$ direction with initia
3	QQ	Single	1.00	1	5	Yes	1	The figure is a section of a conducting rod of radius
4	QQ	Single	1.00	1	5	Yes	1	The electric potential as a function of x is shown in
5	QQ	Single	1.00	0	5	Yes	1	An electron moves in the electrical potential created b
6	QQ	Single	1.00	1	5	Yes	1	In the figure at right all the capacitors are 1 μF.
7	QQ	Single	1.00	1	5	Yes	1	In the figure, all resistances are 400 Ω. Find t
8	QQ	Single	1.00	1	5	Yes	1	In the circuit shown, $\mathcal{E}_1 = 10$ V,
9	QQ	Single	1.00	1	5	Yes	1	In the figure, $\mathcal{E} = 6.0$ V, $R_1 = 20$ Ω,
10	QQ	Single	1.00	1	5	Yes	1	A negatively charged particle goes in a circle perpendi
11	QQ	Single	1.00	1	5	Yes	1	Three wires have current either into or out of the page
12	QQ	Single	1.00	0	5	Yes	1	The magnetic field is coming out of the page with magni
13	QQ	Single	1.00	1	5	Yes	1	A 10 μF capacitor has an initial charge of 100
14	QQ	Single	1.00	1	5	Yes	1	A Gaussian cube of edge length 1.0 m is placed in a mag
15	QQ	Single	1.00	1	5	Yes	1	Initially unpolarized light is sent into a system of th
16	QQ	Single	1.00	1	5	Yes	1	Light travels through layers of transparent slabs, as s
17	QQ	Single	1.00	0	5	Yes	1	A small object is located on the central axis of a sphe
18	QQ	Single	1.00	1	5	Yes	1	In the figure, O (the object) stands on the common ce
19	QQ	Single	1.00	0	5	Yes	1	Light of wavelength 500 nm is sent through two narrow s
20	QQ	Single	1.00	0	5	Yes	1	The distance between the sixth minimum and the central